

GSM-R

Global System for Mobile Communications — Railway

Revision Guide — EMR Training / RS523 Issue 2 (December 2025) / GERT8000-TW5 Issue 14

⚠ All rule book content sourced exclusively from RS523 Issue 2 and GERT8000-TW5 Issue 14 ⚠

1. What is GSM-R?

GSM-R stands for Global System for Mobile Communications — Railway. It is the dedicated digital radio communication system used on the GB mainline railway, replacing analogue radio systems. It provides voice and data communication between drivers, signallers, operations controllers, guards and train operator controllers.

1.1 Two Types of GSM-R Call (RS523 §5.1)

Call Type	Description
Group Calls	Calls made to multiple users simultaneously. Includes: Railway Emergency Group Call (REC), Broadcast area calls, Shunting group calls.
Point-to-Point Calls	Direct calls between two users. Includes: Normal calls, Urgent calls, DSD alarm calls, PA calls, Berth-triggered broadcast calls.

2. Driver's Control Panel (DCP) — Key Buttons (RS523 §2.1)

Button	Function
REC (Red)	Railway Emergency Group Call — highest priority call.
Urgent (Yellow)	Makes a point-to-point urgent call to the controlling signaller. Overrides all calls except a REC.
ST (Acknowledge)	Acknowledges a safety broadcast call to the signaller after the call has fully terminated. Also clears the screen during a REC.
Standing at Signal (SG)	Sends a text message to tell the signaller the train is standing at a signal at danger.
Call Signaller	Makes a direct call to the signaller.
TOC Phonebook	Accesses the train operator's phonebook.
Registration/Turn On	Enables registration and deregistration. Turns DCP on when pressed for 2 seconds.
Accept (✓)	Answers calls, accepts user entries, and triggers pre-registration manually.
Reject/Cancel (✗)	Rejects/deletes entries, ends a call, clears a fault, or immediately deregisters the cab radio.
Test	Tests the train radio.
Menu	Provides access to the in-built menu.
Volume Up/Down	Increases or decreases loudspeaker volume.
Brightness Left/Right	Adjusts display brightness or scrolls text left/right.

You can switch on the cab radio WITHOUT using a driver's key. Limited functions are available without a key, but REC and echo test calls can still be made. (RS523 §3.1)

3. Registration (RS523 §4)

3.1 What is Registration?

When preparing a train for service, the train radio must always be registered with the GSM-R network. This enables the train reporting number to be recognised, allows the signaller to contact the train radio, and supports correct call routing from the cab radio to the controlling signaller.

The driver must always make sure the CORRECT train reporting number is displayed on the train radio. (RS523 §4.1)

3.2 Registration Code Structure

The registration code is 7 digits:

- First 4 digits = Train's headcode (train reporting number, e.g. 1A26).
- Last 3 digits = Location code — obtained from the signal identification plate, an alias plate at which the train is standing, or a company wild card code (99X).

Each TOC/FOC has its own wild card code unless an alias plate is displayed.

You must NOT register while on the move. If pre-registration is needed, store the code when stationary first. (RS523 §4.4)

3.3 Sharing a Platform or Siding

Where two or more trains share a platform or siding, the driver of the second train must NOT register until the first train has departed AND the signal has returned to danger. (RS523 §2.1 / classroom notes)

3.4 Pre-Registration (RS523 §4.4)

If a train needs to transfer to GSM-R at a later stage (e.g. at a boundary), pre-registration is used. The registration code is stored in the cab radio while stationary and the transfer is made on the move later.

- Pre-registration is accessed via the menu.
- When pre-registering, the company wild card code is used.
- Pre-registration can be triggered manually by pressing the Accept button.
- If pre-registration is needed at a location not normally used, the signaller will normally arrange for drivers to be told in advance.

3.5 Pending Registration (RS523 §4.2)

If GSM-R network coverage is not available at the registration point, pending registration is used. The radio stores the registration code and automatically completes registration once coverage is detected.

- The driver enters the headcode and wild card code and presses Accept.
- The message "Searching networks – Please wait" appears until coverage is found.
- The driver must respond to the prompt by pressing Accept when coverage is detected.
- Pending registration can be cancelled at any time by pressing Cancel.

Pending registration is NOT available through the menu — it is initiated by pressing the

Registration button. (RS523 §4.2)

3.6 Registration Problems (RS523 §4.5)

Problem	Action
"Registration Failed" displayed	1. Check code and try again. 2. If second attempt fails, call signaller using phonebook. 3. One further attempt using company wild card code. If "GSM-R GB" shown on screen, train can proceed unregistered. If not shown, treat as defective radio (use TW5 instructions).
Wrong train reporting number displayed	Deregister and re-register using the correct headcode and location code.
"Duplicate" displayed	Registration code is already in use. Contact signaller for new code or alternative wild card. A REC can still be made while unregistered.
Loss of registration during journey	Re-registration required.
Train radio fails to deregister	Attempt to deregister again. If it also fails, report to signaller.
Unregistered train — "GSM-R GB" displayed	Core radio functions still available. Register at a suitable location. REC still works. Use phonebook to contact signaller.
Unregistered — "GSM-R GB" NOT displayed	Treat as defective radio. Carry out TW5 instructions.

3.7 Deregistration (RS523 §9)

The train radio must be correctly deregistered at the end of each journey. Failure to deregister may prevent subsequent registration and can leave the train reporting number stuck on the signaller's terminal (causing a "duplicate" fault).

Method	How
Automatic (end of journey)	Train radio deregisters automatically 10 seconds after the driver's key is removed.
Immediate	Press the Cancel/Reject button — deregisters immediately.
Manual	Press Registration/Turn On button, then press Accept. Successful = blank screen displayed.
Mid-journey (change of headcode)	Manually deregister then re-register via the menu.
Reversing movement	Deregister from the radio in the old leading cab BEFORE registering in the new leading cab.

4. Railway Emergency Group Call (REC) (RS523 §5.2)

4.1 What is a REC?

A REC (Railway Emergency Group Call) is the HIGHEST PRIORITY call on the GSM-R network. It can be made from any terminal or train radio connected to the network and is broadcast to the cell where it was made and adjacent cells.

- Used when an emergency affects more than one line or more than one train.

- If a REC made from a train becomes disconnected, the call stays open until closed by the lead signaller.
- A REC CANNOT be transferred to a third party.

4.2 Receiving a REC — Driver Actions

When a REC is received:

- The train radio sounds: "Emergency, Emergency, Emergency".
- The display shows: "STOP EMERGENCY".
- Any calls already in progress are automatically disconnected.
- STOP THE TRAIN and listen to the message.
- Pressing the ST button stops the display flashing during the REC.
- Use the PTT (Press to Talk) button when speaking; release to listen.
- Good practice: do not speak unless requested by the lead signaller, or if you have important information to add.

A REC is NOT completed until the lead signaller states: "End of railway emergency group call." (RS523 §5.2.2)

Once the REC has ended, any train NOT instructed to remain at a stand can proceed — but ONLY if the driver is certain the train is not affected by the incident.

4.3 PTT Override (RS523 §5.2.3)

If a driver holds the PTT button the lead signaller cannot be heard. The lead signaller can override the driver's PTT by pressing the override button on the fixed terminal.

4.4 Accidental REC (RS523 §5.2.5)

If a REC is initiated accidentally, you must speak to the lead signaller and explain what happened — to avoid extensive delays while the origin of the REC is determined. (RS523 §5.2.5)

5. Urgent Calls (RS523 §5.3)

An urgent call is prioritised by the GSM-R system and overrides all calls except a REC. Used when there is an incident but other trains in the area do NOT need to be stopped immediately.

- The yellow urgent button makes a point-to-point call to the signaller controlling the area.
- If the signaller calls the train urgently, the call is automatically answered after two ring cycles.
- On the signaller's panel, an urgent call causes the active call button to flash yellow.

Urgent calls override all calls except a REC. (RS523 §5.3)

6. Operational Text Messages (RS523 §7)

Message	Who sends it	Meaning / Driver Action
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Standing at Signal (SG)	Driver → Signaller	Used when standing at a signal at danger and not immediately obvious why the train has been brought to a stand. Signaller responds by clearing the signal, sending a "Wait" message, or making a voice call.
Contact Signaller	Signaller → Driver	First method the signaller normally uses to contact the driver. Driver replies ONLY when it is safe to do so by pressing Accept.
Contact Train Operator Control	TOC → Driver	Train operator wishes to speak to the driver. Driver replies when safe by pressing Accept.
Wait	Signaller → Driver	Sent in response to the "Standing at Signal" message. Driver waits for further instructions or clearance.
Acknowledge (ST)	Driver → Signaller	Driver acknowledges a safety broadcast call by pressing ST. Signaller receives confirmation message.

7. Broadcast Calls (RS523 §8)

7.1 Types of Broadcast Call

Type	How it works
Berth-triggered broadcast	Pre-recorded message activated when the train enters a specific berth. Allows the signaller to caution a train without bringing it to a stand. Can be advisory or safety (acknowledged).
Cell-based group broadcast	Sent to all trains within a pre-defined area. Can be pre-recorded or live. Advisory calls only — acknowledged (safety) calls are NOT used with cell-based broadcasts.

7.2 Advisory vs Safety Broadcast Calls

Category	Purpose	Acknowledgement required?
Advisory	To give drivers information — e.g. congestion, blanket speed restrictions, delays.	No — not required.
Safety (Acknowledged)	To pass safety-critical information requiring the driver to confirm understanding.	YES — driver must press ST after call terminates.

7.3 Scenarios for Safety Broadcast Calls

Acknowledged (safety) broadcast calls are used for:

- Missing or obscured Temporary Speed Restriction (TSR) board.
- Poor/reportable railhead conditions.
- Animals on or near the line (NOT in tunnels).
- Defective Emergency Indicators.
- Unusual events (NOT track or signalling related).

7.4 Acknowledging a Safety Broadcast — Critical Rule

Broadcast calls are answered automatically by the train radio (no action needed to receive them). After listening to the full message:

- Wait for the call to be fully terminated — the screen must return to normal display.
- ONLY then press the ST button to acknowledge.

CRITICAL: If you press ST before the broadcast has fully terminated, the acknowledgement will FAIL and the signaller will not clear the signal ahead. There may be 1–2 seconds between the signaller ending the message and the system terminating the call. Wait for the screen to return to normal. (RS523 §8.3)

8. DSD Alarm and PA Calls (RS523 §6)

8.1 DSD Alarm Call (RS523 §6.1)

The train radio sends a DSD alarm message to the signaller after ONE MINUTE if:

- The master switch is in the "forward" or "reverse" position, AND
- No action is taken by the driver to reset the DSD.

If a DSD alarm is triggered by mistake, the driver can contact the signaller and explain what happened.

8.2 PA Calls (RS523 §6.2)

If the signaller cannot contact the driver and believes they may be incapacitated (DOO train), the signaller may make a PA announcement over the train radio loudspeaker to seek assistance. If able to respond, the driver can make a point-to-point call after the PA announcement has been terminated.

9. Short Dial Codes (RS523 §5.4)

Code	Connects to
1200	Controlling Signaller
1300	Operations Control
1400	Electrical Control Operator (ECO)
1900	Echo call test (dial to test if radio can make and receive calls)

Short dial codes only work when the train radio is correctly registered. For unregistered trains, use the phonebook.

You can use the cab radio to call the ECO (short dial 1400). You should only make a

999/112 emergency call as a LAST RESORT. (RS523)

10. Phonebooks and Misrouted Calls (RS523 §5.5 & §5.7)

10.1 Phonebooks

There are two phonebooks on the cab radio:

- Network Rail phonebook — details of signallers, operations controllers and ECOs.
- Train operator's (TOC) phonebook — contacts specific to the train operator.

Entries are listed alphabetically. Search by scrolling or entering the first three letters of the location via the keypad. Used when the train is unregistered or when contacting a signaller outside the registered area.

10.2 Misrouted Calls (RS523 §5.7)

The train radio does NOT indicate network problems that may prevent calls reaching the correct signaller. The driver must always confirm they are communicating with the CORRECT signaller using normal communication protocols.

If a driver receives a broadcast about network problems and needs to contact the signaller, they use the Network Rail phonebook or dial the number from the Periodical Operating Notice (PON).

Always confirm the identity of the signaller once communication is established. Do not assume the call has routed correctly. (RS523 §4.5.1 / classroom notes)

11. Faults (RS523 §10)

11.1 Service-Affecting vs Non-Service-Affecting Failures (RS523 §10.2)

Type	What it means	Action
Service-affecting failure	Train radio unable to enter service. Indicated by: Failure XX (01–07), "Radio Failure", "Cab Radio Flt", "EPROM/RAM Flt", or blank screen.	Report to signaller immediately using alternative communication. Carry out TW5 defective radio instructions.
Non-service affecting failure	Indicated by "Warning" followed by a number. Radio works but may have limited functionality.	Train can enter service without restriction. Driver advises train operator's control before departure.

As long as the train radio can make and receive a REC, the radio will generally NOT be considered defective. (RS523 §10.2.1)

11.2 Network Coverage Issues (RS523 §10.3)

If coverage is temporarily lost, the display shows "Searching networks – Please wait" and a warning tone sounds. If coverage is not regained after passing through the affected area, treat the radio as defective and carry out TW5 instructions.

If the radio connects to the wrong network, or displays "Network?", select the correct network manually: Menu → Settings → Select Network → GSM-R GB.

11.3 Echo Call Test (RS523 \$10.5)

To check the radio can make and receive calls: dial 1900 using the keypad, record a short message, wait a few seconds for it to be played back. If the message is not played back, inform the signaller using alternative communication.

11.4 Silent Radio (RS523 \$10.8)

Silent radio occurs if power is removed and restored within 10 seconds. Symptoms: no REC introduction message, no double-beep tone during registration, no single-beep when pressing Cancel. Prevent by leaving at least a 10-second gap when cycling power.

To resolve: place handset in cradle and press volume button three times. A failed action tone on Cancel confirms the issue is resolved. If not, remove power for 3 minutes and repeat.

11.5 Defective Radio — TW5 Rules (GERT8000-TW5 \$26)

The radio rules from GERT8000-TW5 Issue 14 apply when the cab radio is defective:

Starting from a maintenance depot (\$26.1)

You must NOT allow a train to start a journey with a defective radio in any cab which is required to be driven from at any time during its planned working. (TW5 \$26.1)

Starting from somewhere other than a maintenance depot (\$26.2)

You must NOT allow a train to start with a defective radio in the driving cab UNLESS one of the following applies:

- Operative transportable or portable GSM-R radio has been provided in the driving cab.
- Permission given to proceed to a location where a portable/transportable radio will be provided.
- A competent person is in an alternative cab with an operative radio (with a direct means of communication between them and the driver).
- Permission given to proceed to a location where such a competent person will be provided.
- Permission given to complete the journey without an operative radio.
- The defective radio is not in the cab from which the train will be driven on this journey.
- You have been told there is a radio NETWORK FAILURE.

Radio network failure speed restriction: if told of a network failure you may be instructed not to exceed 100 mph (160 km/h) or 60 mph (100 km/h) through the affected area. (TW5 \$26.2)

Radio becomes defective during a journey (\$26.3)

If the cab radio becomes defective during a journey:

- Tell the signaller as soon as possible — stop the train specially if necessary.
- Do not move the train until instructed to do so.
- Carry out the instructions given.

The train can continue if one of the same conditions in §26.2 is met.

12. Keeping Registration / Key Removal (RS523 §2.1 / classroom notes)

If you want to keep your registration when removing your key, you have 10 seconds to retain the registration by pressing the tick (Accept) button. (RS523 classroom notes)

13. Questions and Answers

Q1. What does GSM-R stand for?

A1. Global System for Mobile Communications — Railway.

Q2. What are the two main types of GSM-R call?

A2. Group calls (REC, broadcast, shunting group) and point-to-point calls (normal, urgent, DSD alarm, PA, berth-triggered broadcast).

Q3. What is the structure of a registration code?

A3. 7 digits: first 4 = train headcode; last 3 = location code (from signal ID plate, alias plate, or company wild card code 99X).

Q4. Can you register whilst on the move?

A4. No. You must NOT register whilst on the move. You may store the code (pre-registration) when stationary and transfer on the move.

Q5. What is pending registration?

A5. Used when GSM-R coverage is not available. The registration code is stored and the radio automatically completes registration when coverage is detected. The driver must press Accept when prompted.

Q6. Can pending registration be cancelled?

A6. Yes — by pressing the Cancel button at any time.

Q7. The first registration attempt fails. What is the procedure?

A7. (1) Check code and attempt again. (2) If second attempt fails, call signaller using phonebook. (3) Try once more using the company wild card code. If 'GSM-R GB' is displayed, proceed unregistered. If not displayed, treat as defective radio.

Q8. 'Duplicate' is displayed on the cab radio. What does this mean?

A8. The registration code is already in use by another train radio. Contact the signaller for a new code or alternative wild card.

Q9. Two trains are sharing a platform. When can the second driver register?

A9. Only after the first train has departed AND the signal has returned to danger.

Q10. What happens if you do NOT correctly deregister at the end of a journey?

A10. Subsequent registration attempts may fail. The train reporting number may remain on the signaller's terminal. A 'duplicate' fault may be displayed.

Q11. How does the train radio automatically deregister?

A11. Automatically 10 seconds after the driver's key is removed. Or immediately by pressing Cancel. Or manually via Registration button then Accept.

Q12. What is a REC and when is it used?

A12. Railway Emergency Group Call — the highest priority call. Used when an emergency affects more than one line or more than one train.

Q13. When the driver receives a REC, what must they do?

A13. Stop the train and listen to the message. The radio announces 'Emergency, Emergency, Emergency' and displays 'STOP EMERGENCY'.

Q14. How does a driver know a REC has ended?

A14. The lead signaller states: 'End of railway emergency group call.' The REC is NOT complete until this phrase has been heard.

Q15. After a REC ends, when can a train proceed?

A15. When the driver is certain the train is not affected by the incident AND the driver has not been instructed to remain at a stand.

Q16. What happens if a REC is accidentally initiated?

A16. You must speak to the lead signaller and explain what happened, to avoid extensive delays while the origin is investigated.

Q17. What is an urgent call and how does it differ from a REC?

A17. An urgent call is a point-to-point call to the controlling signaller, prioritised above all calls except a REC. Used when there is an incident but other trains don't need to be stopped immediately.

Q18. What does the ST button do?

A18. It acknowledges an acknowledged (safety) broadcast call to the signaller. It also stops the display flashing during a REC.

Q19. What is the critical rule when pressing the ST button after a safety broadcast?

A19. Only press ST once the broadcast has fully terminated and the screen has returned to normal. If pressed too early, the acknowledgement fails and the signaller will NOT clear the signal ahead.

Q20. Why might there be a 1-2 second delay before pressing ST?

A20. There can be 1-2 seconds between the signaller ending the message and the system terminating the call. Wait for the screen to return to normal to be sure.

Q21. For what scenarios are safety (acknowledged) broadcast calls used?

A21. Missing/obscured TSR board, poor railhead conditions, animals on/near the line (not in tunnels), defective emergency indicators, unusual events (not track or signalling).

Q22. What is the difference between an advisory and a safety broadcast call?

A22. Advisory calls give information (no acknowledgement required). Safety calls are safety-critical and MUST be acknowledged by the driver pressing ST.

Q23. What is the 'Standing at Signal' (SG) message used for?

A23. Used by the driver to tell the signaller the train is standing at a signal at danger and it is not immediately obvious why.

Q24. What does the 'Contact Signaller' message mean for the driver?

A24. The signaller wishes to speak to the driver. The driver replies ONLY when it is safe to do so by pressing Accept.

Q25. How long before the DSD alarm is sent to the signaller?

A25. One minute after no action is taken to reset the DSD (with master switch in forward or reverse position).

Q26. What short dial code connects to the controlling signaller?

A26. 1200.

Q27. What short dial code connects to Operations Control?

A27. 1300.

Q28. What short dial code connects to the ECO?

A28. 1400.

Q29. How do you perform an echo call test, and why?

A29. Dial 1900, record a short message, wait for playback. Used to check the radio can make and receive calls. If not played back, inform the signaller by alternative communication.

Q30. What are the symptoms of 'silent radio'?

A30. No REC introduction message, no double-beep during registration, no single-beep when pressing Cancel.

Q31. How do you prevent silent radio?

A31. Leave at least a 10-second gap between removing and restoring power on the radio.

Q32. What is a service-affecting radio failure? Give examples.

A32. A failure that means the radio cannot enter service: Failure XX (01–07), 'Radio Failure', 'Cab Radio Flt', 'EPROM/RAM Flt', or blank screen. Carry out TW5 defective radio instructions.

Q33. When can a train start a journey from a maintenance depot with a defective radio?

A33. It cannot. You must NOT allow a train to start a journey from a maintenance depot with a defective radio in any cab required to be driven from.

Q34. A radio becomes defective DURING a journey. What must the driver do first?

A34. Tell the signaller as soon as possible — stop the train specially if necessary. Do not move until instructed.

Q35. What speed restriction may apply when passing through a radio NETWORK FAILURE area?

A35. You may be told not to exceed 100 mph (160 km/h) or 60 mph (100 km/h) through the affected area. You must not exceed the speed you have been told.

Q36. If you want to retain your registration when removing your key, how long do you have?

A36. 10 seconds — press the tick (Accept) button within that time to keep the registration.

Q37. A misrouted call is received. What must the driver always check?

A37. Always confirm they are communicating with the CORRECT signaller using normal communication protocols. Never assume the call has routed correctly.

Q38. A REC cannot be transferred — TRUE or FALSE?

A38. TRUE. A REC cannot be transferred to a third party. (RS523 §5.10)